Determination of Breakdown Voltage and Breakdown Strength In Insulating Oils & LCOE High Voltage Laboratory

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What is transformer oil ?

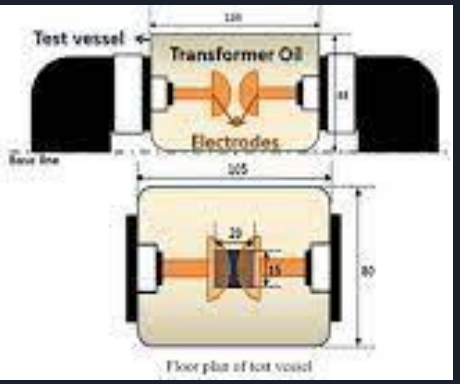
Transformer oil or insulating oil is an oil that is stable at high temperatures and has excellent electrical insulating properties. It is used in oil-filled transformers, some types of high-voltage capacitors and some types of high-voltage switches and circuit breakers. Oil used in transformers acts as a coolant for the windings, as an insulant to prevent arcing between parts of the transformer circuits.

	Oil	Air	SF_6 / Vacuum
Security	Can explode, burn, increasing pressure inside the tank is risky.	Can be impacted by external factors, while breaking hot and ionized gas is emitted.	No impact due to external factors, no risk of explosion or flamability.
Size	Large	Large	Small
Maintenance	Frequent, change of oil	Change of contacts, control of mechanical parts	No maintenance of breaker parts, minimal oiling for mechanical parts
Sensitivity to environmental factors	Sensitive	Sensitive	Not sensitive, breaking takes place in closed space
Recovery time	Lower breaking capability in case of increased pressure	Lower breaking capability in cae of hot air emission	Fast dielectric recovery, can break consecutively.
Durability	Medium	Medium	Excellent

Breakdown Voltage and Dielectric Breakdown

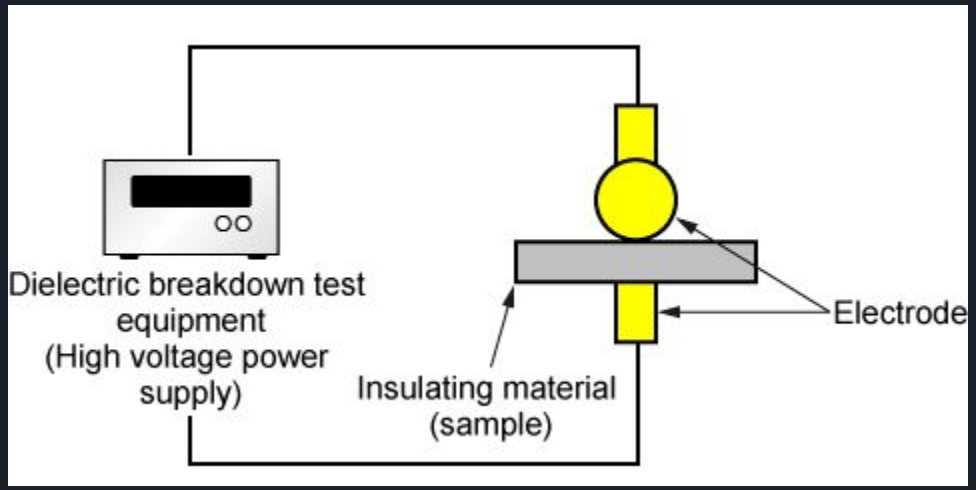
When a voltage is applied to the insulating oil, as the voltage increases, the current through the oil increases sharply, causing it to completely lose its inherent insulating properties and become a conductor. This occurrence is called as breakdown of the insulating oil. The critical voltage value at which the insulating oil breaks down is called as the breakdown voltage or dielectric breakdown. The electric field strength at this time, called as the dielectric strength of the oil, indicates the ability of the insulating oil to resist the electric field.





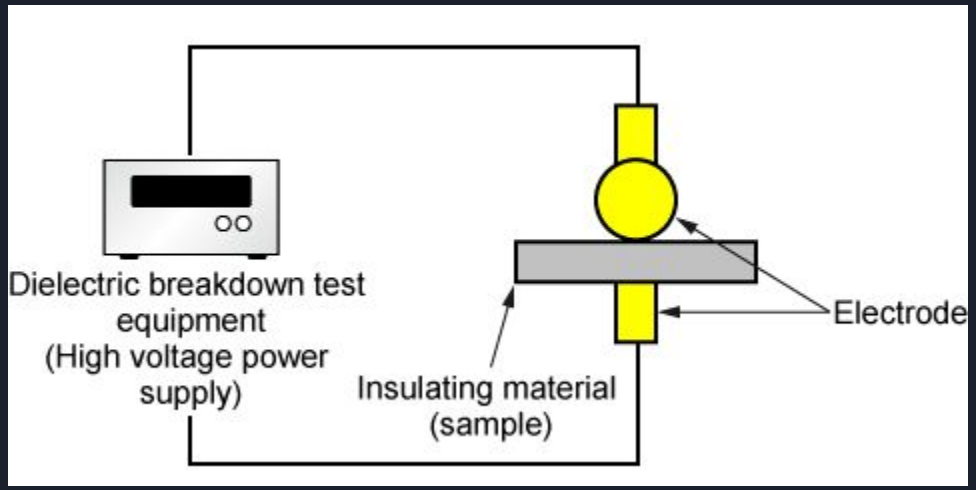
What is oil dielectric breakdown voltage testing?

Put simply, a dielectric breakdown voltage test is a measure of the electrical stress that an insulating oil can withstand without breakdown. The test is performed using a test vessel that has two electrodes mounted in it, with a gap between them. A sample of the oil to be tested is put into the vessel and an ac voltage is applied to the electrodes. This voltage is increased until the oil breaks down – that is, until a spark passes between the electrodes. The test voltage is then immediately turned off.



What is oil dielectric breakdown voltage testing?

The voltage at which breakdown occurred is the test result, and is typically evaluated by comparing it with guidelines set out in various standards, or in the oil manufacturer's specifications. The exact method of performing the test is determined by the standard that is being used, as will be explained later. The standard typically defines parameters such as the size and shape of the electrodes, the gap between them, the rate at which the test voltage is increased, how many times the test is repeated and whether or not the oil is stirred during the test.



Calculations After the Test

After recorded 5 breakdown voltage value, we do average and standard deviation of results:

$$\bar{X} = 1/5 \sum_{i=1}^{5} X_i \quad \text{and} \quad s = \sqrt{1/4[\sum_{i=1}^{5} X_i^2 - 5\bar{X}^2]}$$

where:

$\bar{X}_i$ = mean of the five individual values,

X_i = ith breakdown voltage, and

s = standard deviation.

The Accuracy of the Test

According to the results we can understand the accuracy of the test with 2 ways:

1) If the ratio of $s/\bar{X}$ exceeds 0.1, it is probable that test result is excessive.

2) The range of the five breakdown (Maximum breakdown voltage minus minimum breakdown voltage) times 3 value is greater than the next to the lowest breakdown, it is probable that test result is excessive.

What are the test standards and their differences?

There are many test standards for insulating liquids, but they are derivatives of three main standards. Two of these are from ASTM International (USA) and the other is from the IEC (Europe).

The 3 main standards will be given in the next page. But in addition to those standards, there is also a Japanese standard not based on these ASTM or IEC standards.

Which Standards Are Applied In This Test?

3 important standards that scientists follow:

-ASTM D1816 – Standard Test Method for Dielectric Breakdown Voltage of Insulating Oils of Petroleum Origin Using VDE Electrodes

-ASTM D877 – Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes

-IEC 60156 Insulating Liquids – Determination of the Breakdown Voltage at Power Frequencies





ASTM D1816 – Standard Test Method for Dielectric Breakdown Voltage of Insulating Oils of Petroleum Origin Using VDE Electrodes

-ASTM D1816 covers dielectric breakdown voltage of insulating oils of petroleum origin.

-Suitable for oils having viscosities of more than 19 (mm2/s) at 40 degrees Celsius.

-This test is not recommended for oil received from vendors in tank cars, tank trucks or drums.

-Electrodes shall be polished brass spherically-capped electrodes of VDE(Verband Deutscher Elektrotechniker, Specification 0370). Electrodes shall be 36mm diameter, 25mm radius and 13mm thick.

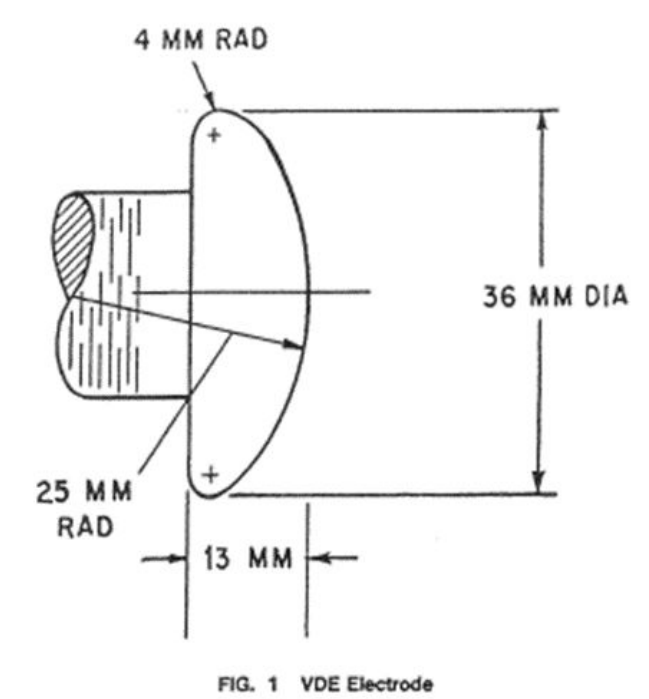


FIG. 1 VDE Electrode

-Electrode gap is 2±0.03-mm at 0.9L test sample and 1±0.03-mm at 0.5L test sample.

-Transformer's AC signal 45-60Hz test voltage shall differ by no more than ±5%.

-The rate of voltage rise shall be 0.5 kV/s ±20% starting from 0 V.

ASTM D877 – Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes

-ASTM D877 Covers dielectric breakdown voltage of insulating oils of petroleum oils, hydrocarbons and askarels commonly used at transformers, oil circuit breakers, etc.

- Suitable for oils having viscosities of more than 900 (mm2/s) at 40 degrees Celsius.

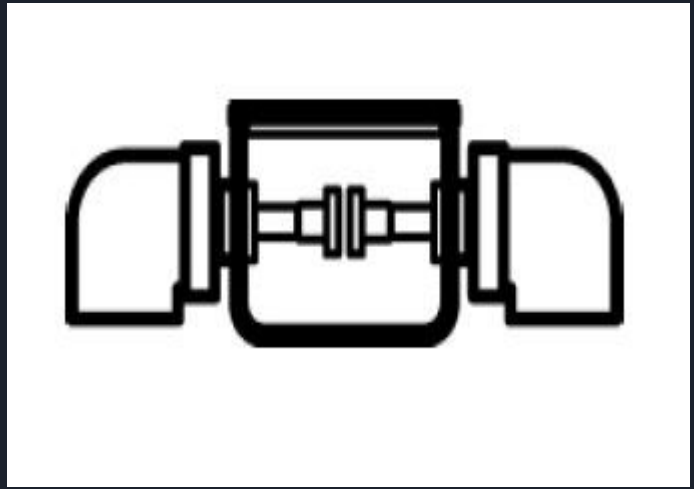
- This test is recommended for on unprocessed insulating oil received from vendors in tank cars, tank trucks or drums.

-Electrodes shall be polished brass disks 25 mm diameter and at least 3mm thick with square edges.

-Electrode gap is 2.5±0.01-mm.

-Transformer’s AC signal 45-60Hz test voltage shall differ by no more than ±5%.

-The rate of voltage rise shall be 3 kV/s ±20% starting from 0 V.



IEC 60156 Insulating Liquids – Determination of the Breakdown Voltage at Power Frequencies

-Test is appropriate for both unused liquids and establishing the condition of samples taken in monitoring and maintenance of equipment.

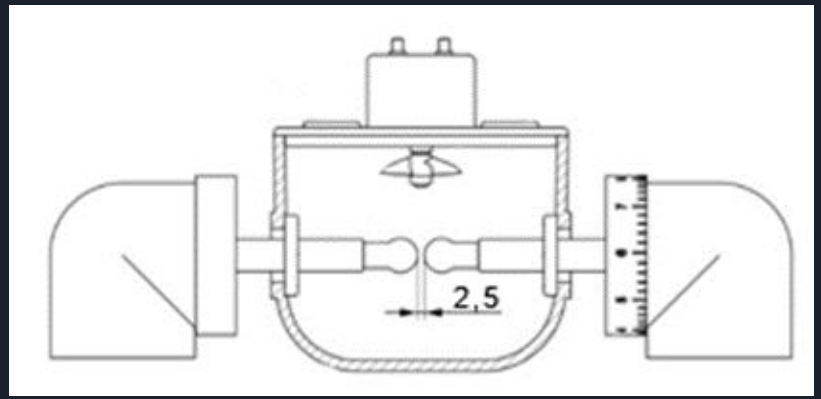
-The method applies to all types of insulating liquids of nominal viscosity up to 350 mm2/s at 40 degrees Celsius.

-The electrodes shall be made either of brass, bronze or austenitic stainless steel. They shall be polished and, in shape, either spherical (12.5 mm to 13.0 mm diameter) or in partially spherical shape (25 mm ± 0.25 mm radius).

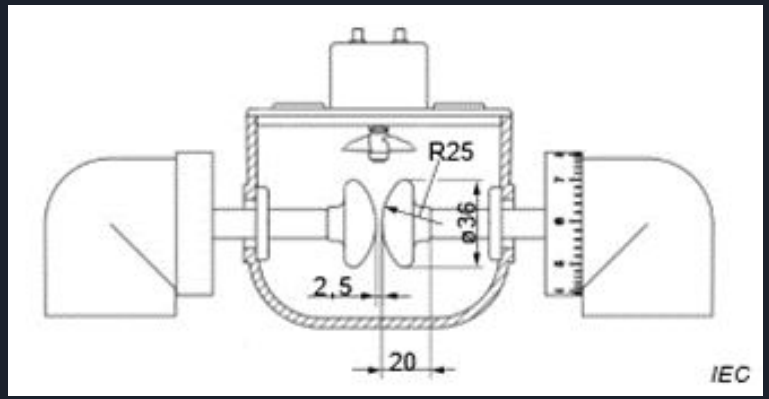
-The gap between the electrodes shall be 2.50 mm ± 0.05 mm.

-Transformer’s AC signal 48-62 Hz test voltage shall differ by no more than ±2%.

-The rate of voltage rise shall be 2 kV/s ±0.2 kV/s starting from 0 V.



Spherical Configuration



Partially Spherical Configuration

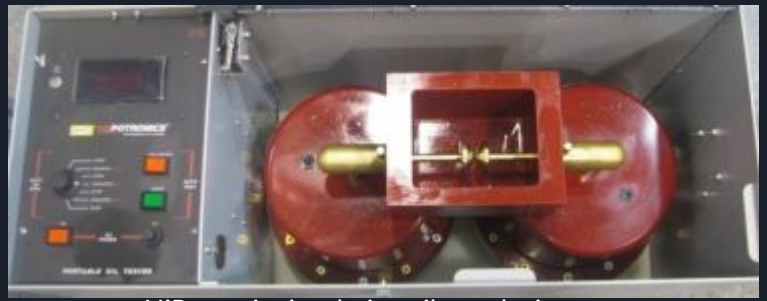
Breakdown Strength Analysis of Different Insulating Oils Due To Different Standards

This test applied by Turkish scientists at 2013 and aiming to observe the result when different standards applied.

Electrode gaps are constant 2.5mm.

Two different electrode types (cylindrical and spherical).

Two different oil type (aged and non-aged)



HiPotronics insulation oil test device



Results:

Non-aged oil spherical electrode

Voltage Rising Rate (kV/s)	Electrode Gap (mm)	$V_{Breakdown}$ at 2,5 mm (kV)	$E_{Breakdown}$ strenght (kV/mm)
0.5	2.5	60.37	34.42
2	2.5	70.20	40.01
3	2.5	74.42	42.42

Non-aged oil cylindrical electrode

Voltage Rising Rate (kV/s)	Electrode Gap (mm)	$V_{Breakdown}$ at 2,5 mm (kV)	$E_{Breakdown}$ strenght (kV/mm)
0.5	2.5	51.17	20.47
2	2.5	57.65	23.06
3	2.5	56.28	22.51

Aged oil spherical electrode

Voltage Rising Rate (kV/s)	Electrode Gap (mm)	$V_{Breakdown}$ at 2,5 mm (kV)	$E_{Breakdown}$ strenght (kV/mm)
0.5	2.5	44,10	25.14
2	2.5	59,88	34.13
3	2.5	55,30	31.52

Aged oil cylindrical electrode

Voltage Rising Rate (kV/s)	Electrode Gap (mm)	$V_{Breakdown}$ at 2,5 mm (kV)	$E_{Breakdown}$ strenght (kV/mm)
0.5	2.5	37.42	14.97
2	2.5	45.37	18.14
3	2.5	47.10	18.84

LCOE High Voltage Technological Centre

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Brief Description of LCOE



The LCOE High Voltage Technological Centre is a prominent organization in Spain that specializes in independent testing and metrology services for the electrical sector. Their expertise lies in high voltage testing, research, electrical metrology, and calibration.

LCOE is an ISO 17025 accredited testing and calibration laboratory, recognized by ENAC (Spanish National Accreditation Entity). Additionally, they are currently applying for STL (Short-circuit Testing Liaison) accreditation through the Spanish association of high power laboratories, AELP.

Due to its well-established structure, experienced staff, state-of-the-art facilities, and numerous national and international accolades, LCOE is considered a global authority in the fields of High Voltage and Metrology.

History of LCOE

- **1949:** LCOE creation devoted to electrical tests and energy meter verification.
- **1952-1957:** Starting of the High Voltage and calibration Laboratory.
- **1954:** Decree of Electrical verifications of 12th of March (article 42) recognizes LCOE as official laboratory for energy meter verification.
- **1992:** Creation of the Foundation, FFII, in which the LCOE was integrated.
- **1993:** RELE (ENAC) accreditation for High Voltage testing, according to EN 45001.
- **1995:** RELE (ENAC) accreditation for calibration according to EN 45001.
- **2001:** New Laboratory facilities in Getafe, for High current impulse tests and metrology.
- **2001:** LCOE designated as the national laboratory for developing and maintenance of national standards in HV, according to RD 346/2001.

- **2002:** ENAC Accreditation according to ISO/EN 17025 for testing and calibration laboratories.
- **2004:** LCOE recognized by EURAMET and BIPM as DI with CMC´s for High Voltage quantities.
- **2011:** Movement from the Polytechnic University facilities to the new installations in TECNOGETAFE.
- **2012:** First participation of LCOE in Euramet Projects.
- **2014:** Creation of CTAT, High Voltage Technological Center.
- **2015:** Member of AELP (Spanish Association of Power Laboratories) and applicant of STL since 2016.
- **2019:** New CMC´s recognized by the BIPM, including High voltage and high current quantities.

Which tests are being conducted in the lab?



The lab supports various types of HV tests ranging from Impulse Tests to Discharge tests. The HV lab equipment is capable of generating frequency power up to 1 MV and impulse voltages up to 2.4MV for lightning and 1.6MV for switching impulses. The lab has two halls, one for High Voltage (1000m^2) and the other for Medium Voltage (377m^2). The medium Voltage Hall contains five testing cells and those are capable of performing tests up to 400kV.

The laboratory also contains mobile testing platforms for on-site tests. These equipments have voltage ranges between 36kV to 680kV and they are capable of testing currents up to 83A.

Calibration Tests

There are also HV and LV calibration tests conducted at LCOE. These calibration tests are conducted on equipment pieces such as partial discharge meters, voltage dividers, and transformer radiometers.

They also provide on-site AC and DC calibration equipment. These calibration equipment are capable of generating up to 2500kV AC and 1000kV DC



Aircraft Lightning Strike Tests

Another type of test that can be conducted in the lab is Aircraft lightning strike tests. LCOE is collaborating with major aircraft producers for more than 20 years. High voltage tests and currents tests are conducted on precious aircraft parts such as winglets and rotor blades. These parts has to be tested carefully since they might cause part loss on a lighting strike.

There are also voltage and current transformers tests on the lab. Alongside all these power system equipments, personal power protection equipments are also tested on the lab.



SYSTEMS	RATINGS
DC Measuring systems	Up to 200 kV
Partial discharge instruments	From 0.5 to 2000 pC
Partial discharge calibrators	From 0.5 to 100 000 pC
Megohm meters	Up to 15 kV
Capacitance/tan δ bridges	From tan δ ≥ 10-5 pu
DC Measuring systems	Up to 200 kV
AC Measuring systems	Up to 500 kV
LI / SI Measuring systems	Up to 800 kV
Voltage transformers	Up to 500 kV
Current transformers	Up to 10 kA
High voltage capacitors	Up to 200 kV

Table 1: HV calibration tests and their ratings that can be initiated at LCOE.

MAGNITUDE	RANGE
Frequency	10 mHz ≤ f ≤ 100 MHz
Period	2 ns ≤ T ≤ 5 s
Time interval	40 ms ≤ Δt ≤ 100 s
Bandwidth	Up to 1 GHz

Table 2: Time & Freq values of LV measuring equipment.

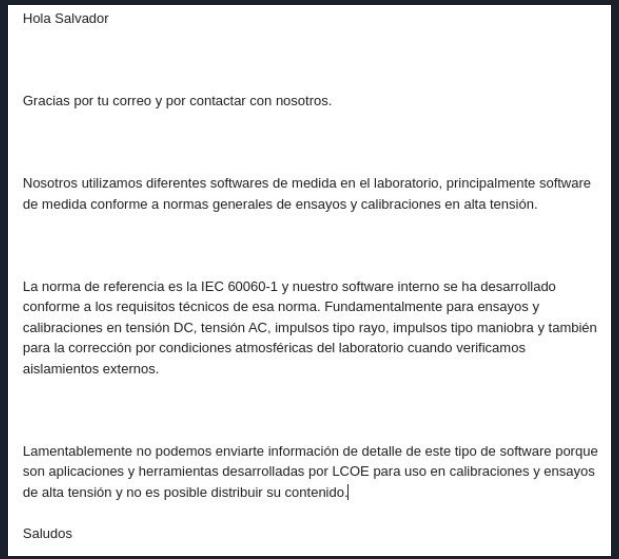
Which standards are followed during the tests?

Every Lab has to follow several standards created by a well-known organization before initialing the tests. LCOE follows a bunch of regulations and standards before initiating and completing the tests. The standards and regulations can be summarized as follows:

- IEC 61869, IEEE C57.13 -> High & Medium Voltage transformers
- IEC 61869-5, ANSI C 98.1 -> Capacitive Voltage transformers.
- IEC 61869-4 -> Combined Transformers
- IEC 61869-6, IEC 61869-10, IEC 61869-11 -> Low power transformers (Both Current and Voltage).
- IEC 61869-14, IEC 61869-15 -> For DC instrument transformers.
- IEC 60358-1, IEC 60358-2, ANSI C93.1 -> HV Capacitors and Coupling Capacitors.
- IEC 60076-1, (EU) No 548/2014, IEC 60076-2, IEC 60076-11 and IEEE C57.12.90 -> Power transformers.
- EUROCAE ED105 and ED84 and RTCA-DO-160 -> Aircraft lightning tests
- EC 60099-4, IEC 61643, and IEEE C62.11 -> HV & LV surge arresters for grids.
- IEC 60383-1 -> Ceramic and glass insulators.
- IEC 60660 -> Tests on indoor post insulators of organic material.
- IEC 60502, IEC 60840, IEC 62067, HD 629.1, AND IEC 61238-1 -> For cable testing

Software, Hardware and Special Equipments Used in the Lab

The LCOE does not provide any information about the software and the special equipment used throughout their tests. Our team member Salvador sent an email in Spanish about these equipments and softwares to gather more info. We believe they have created their own HV software.



Hola Salvador

Gracias por tu correo y por contactar con nosotros.

Nosotros utilizamos diferentes softwares de medida en el laboratorio, principalmente software de medida conforme a normas generales de ensayos y calibraciones en alta tensión.

La norma de referencia es la IEC 60060-1 y nuestro software interno se ha desarrollado conforme a los requisitos técnicos de esa norma. Fundamentalmente para ensayos y calibraciones en tensión DC, tensión AC, impulsos tipo rayo, impulsos tipo maniobra y también para la corrección por condiciones atmosféricas del laboratorio cuando verificamos aislamientos externos.

Lamentablemente no podemos enviarte información de detalle de este tipo de software porque son aplicaciones y herramientas desarrolladas por LCOE para uso en calibraciones y ensayos de alta tensión y no es posible distribuir su contenido.

Saludos

However they are not allowed to publish the software they created. They provided us with the information of they use several softwares and their reference standard is IEC 60060-1.

References

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